FORGE GROWTH

Installation Guide

Every way to install it — a server from published images or the download bundle, a server from source, a Mac or Windows laptop — then attaching a domain and connecting WhatsApp and Instagram. Every command can be copied as-is.

Repository	github.com/Forgemind-git/ForgeGrowth-OSS
Release	v3.1.1

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Which sections do I need?

Situation	Read
Installing on a customer's server	1 → 2a → 3 and/or 4
Server already runs Traefik, nginx or another site	1 → 2b → 3 and/or 4
You intend to change the code	5
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> **One rule that saves the most time**
>
> The directory you are standing in is what selects the install. Every command in this guide is run from the install directory, because one machine can hold several installs and they are told apart by nothing else.

0 Before you start

Everything runs in containers. You do not install Node, PostgreSQL, Redis or MinIO on the machine.

Requirement	Detail
Docker	with the Compose v2 plugin. `docker compose version` must work — note the space, not `docker-compose`
Shell	Linux, macOS, or Windows through WSL2 / Git Bash. There is no PowerShell installer
RAM	2 GB only if building from source (section 5). The published-image route has no build step
Disk	~3 GB. The first download is about 350 MB
Ports	One for the web UI (8080 by default; the installer moves to the next free one). Ports 80 and 443 as well, if this machine is to obtain its own HTTPS certificate

Check the machine is ready

```
docker compose version
docker run --rm hello-world
```

If the second command prints a greeting, Docker works. If either fails, fix that before continuing — every failure after this point will otherwise be blamed on the installer.

What gets created

The installer makes one directory — `forge-growth` — holding a handful of small files and no source code:

```
forge-growth/
  .env                   your settings and every generated secret
  .forgegrowth-install   which version this install is pinned to
  docker-compose.yml     downloaded; overwritten on every upgrade
  caddy/Caddyfile        only used when this machine handles HTTPS itself
  up.sh  down.sh         start and stop
  install.sh             a copy of the installer, for upgrades
```

Your data is not in that folder. It lives in Docker volumes, which is why `./down.sh` is safe and `docker compose down -v` destroys everything:

```
docker volume ls | grep forgegrowth
```

1 Server, from published images

The normal way to install on a customer's machine. Nothing is cloned, nothing is compiled, and every secret is generated for you.

The machine owns ports 80 and 443, and DNS already points at it

```
bash -c "$(curl -fsSL https://raw.githubusercontent.com/Forgemind-git/ForgeGrowth-OSS/
v3.1.1/scripts/install.sh)" -- \
  --version v3.1.1 --domain crm.example.com --yes
```

That is the whole install. It pulls the images, generates every secret, obtains a Let's Encrypt certificate, starts the stack, fetches the public URL to prove it answers, and prints the admin password once. `--yes` means it asks nothing.

Both halves of the pin matter: the URL fixes which script runs, and `--version` fixes what that script then downloads and pulls. The choice is remembered, so a later bare re-run cannot drift onto a newer build.

Something else already owns 80 and 443 (Traefik, nginx, another site)

```
bash -c "$(curl -fsSL https://raw.githubusercontent.com/Forgemind-git/ForgeGrowth-OSS/
v3.1.1/scripts/install.sh)" -- \
  --version v3.1.1 --url https://crm.example.com --port 8080 --yes
```

Use `--url`, not `--domain`. It tells the installer HTTPS is terminated elsewhere, so it does not start a second web server to fight over the ports. Then follow section 2b.

No domain yet — reach it by IP

```
bash -c "$(curl -fsSL https://raw.githubusercontent.com/Forgemind-git/ForgeGrowth-OSS/
v3.1.1/scripts/install.sh)" -- \
  --version v3.1.1 --url http://203.0.113.10:8080 --yes
```

> **A plain-HTTP address is fine to start with, but it cannot connect WhatsApp or Instagram**
>
> Meta refuses any webhook address that is not `https://`. Add a domain later (section 2) — no reinstall is needed.

Or: the download bundle

For handing an install to someone who will not paste a command from the internet. Download `forge-growth-v3.1.1.zip` from the releases page, unzip it, open the folder for the operating system (`linux`, `macos` or `windows`), then:

```
chmod +x *.sh
./install.sh
```

The installer continues in that folder — what you unzipped is the install — and the bundle stays on its own version. It still needs the internet the first time, to pull the images.

All the flags

Flag	Meaning
`--domain <host>`	Serve HTTPS here, certificate included. Needs ports 80 + 443 free
`--url <origin>`	Public address when something else terminates HTTPS. Include the port if people use one
`--port <n>`	Host port for the UI (default 8080, or the next free one)
`--version <ref>`	Pin the downloaded files and the image tag together. `v3.1.1` and `3.1.1` both work
`--dir <path>`	Install here instead of `./forge-growth`
`--admin-email <addr>`	First admin account (default `admin@<your domain>`)
`--admin-password <pw>`	First admin password (default: generated and printed)
`--tls-email <addr>`	Certificate contact. `internal` self-signs, for a domain with no public DNS
`--source`	Build from a checkout rather than pulling images
`--yes`, `-y`	Never ask. With no address flag: this machine's public address, or localhost

Sign in

The installer prints the URL, the admin email and the password. To read the password again:

```
cd forge-growth
grep '^BOOTSTRAP_ADMIN_PASSWORD=' .env
```

Managing the stack afterwards

Command	Does
`./up.sh`	Start, then verify the public URL really answers
`./down.sh`	Stop. Keeps all data
`./install.sh`	Upgrade in place
`docker compose logs -f backend`	Follow the backend log
`docker compose ls`	List every install on this machine
`docker compose down -v`	Deletes all data. Take a dump first

Use up.sh and down.sh, not docker compose up -d

They perform the only two checks that catch the failures this software actually suffers: a site that returns 404 while every container reports healthy, and a second folder quietly adopting this install's database. Raw Compose skips both.

2 Attaching a domain

Which instructions apply depends on one thing: does this install own ports 80 and 443, or does something else? The installer worked it out and recorded the answer.

```
cd forge-growth
grep '^TLS_MODE=' .env
```

Value	Meaning	Follow
`TLS_MODE=caddy`	This install owns the ports and obtains its own certificate	2a
`TLS_MODE=proxy`	Something else owns them; that thing must route the domain here	2b

2a — This install owns ports 80 and 443

1. Point the domain's A record at the server's IP address, and wait until it resolves.
2. Re-run the installer with the domain. Your data and secrets are kept:

   ```
   cd forge-growth
   ./install.sh --domain crm.example.com
   ```

3. The bundled Caddy obtains the certificate, and the installer confirms it works before reporting success.

> **Behind Cloudflare's orange cloud?**
> The certificate challenge stops at the CDN and never reaches this machine. Switch the record to "DNS only" while the certificate is issued, or use 2b with the CDN supplying the certificate.

2b — Something else owns the ports

Your existing proxy must accept the hostname and forward it to this install's `web` container on port 80. First tell the install its public address, so the API accepts requests from it:

```
cd forge-growth
./install.sh --url https://crm.example.com
```

Traefik, running in Docker on the same host. Create an overlay file outside the install directory — `docker-compose.yml` is re-downloaded on every upgrade, so labels added there are eventually lost:

```
services:
  web:
    networks: [default, proxy_default]
    labels:
      - traefik.enable=true
      - traefik.docker.network=proxy_default
      # ONE service, named explicitly on every router below.
      - traefik.http.services.forgegrowth.loadbalancer.server.port=80
      - "traefik.http.routers.forgegrowth.rule=Host(`crm.example.com`)"
      - traefik.http.routers.forgegrowth.entrypoints=websecure
      - traefik.http.routers.forgegrowth.tls=true
      - traefik.http.routers.forgegrowth.tls.certresolver=myresolver
      - traefik.http.routers.forgegrowth.service=forgegrowth
networks:
  proxy_default:
    external: true
```

Save it as `/opt/forge-growth-proxy.yml`, replace `proxy_default` with the network Traefik is on, and `myresolver` with your certificate resolver. Find the network with:

```
docker inspect <traefik-container> \
  --format '{{range $k,$v := .NetworkSettings.Networks}}{{$k}}{{"\n"}}{{end}}'
```

Wire it in through `.env` — never an exported shell variable — then restart:

```
cd forge-growth
echo 'COMPOSE_FILE=docker-compose.yml:/opt/forge-growth-proxy.yml' >> .env
./up.sh
```

> **Two mistakes here cost an afternoon each**
>
> **One service per router.** Declaring a `loadbalancer.server.port` for each router makes Traefik unable to choose, so it discards every router on that container and each hostname returns 404. **No certresolver behind a CDN.** If the domain is proxied by Cloudflare or similar, the ACME challenge can never succeed, and its retries are rate-limited per account — enough to stop every other domain on the server from renewing. Use `tls=true` with no resolver and let the CDN supply the certificate.

nginx, running outside Docker

```
server {
    listen 443 ssl;
    server_name crm.example.com;
    ssl_certificate     /etc/letsencrypt/live/crm.example.com/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/crm.example.com/privkey.pem;
    client_max_body_size 100M;          # media uploads; nginx defaults to 1M
    location / {
        proxy_pass http://127.0.0.1:8080;      # WEB_PORT from .env
        proxy_set_header Host              $host;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_set_header X-Forwarded-Host  $host;
        proxy_set_header X-Forwarded-For   $proxy_add_x_forwarded_for;
        proxy_http_version 1.1;
        proxy_set_header Upgrade    $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_read_timeout 600s;
    }
}
```

2c — Prove it actually works

A page that loads in your browser shows the interface was served. It does not show the API behind it is reachable — and that is the half which breaks sign-in. Check both:

```
curl -s -o /dev/null -w '%{http_code}\n' https://crm.example.com/
200
curl -s https://crm.example.com/api/auth/me
{"error":"Unauthorized"}
```

Result	Meaning
`200`, then `{"error":"Unauthorized"}`	Working. The proxy reaches this install, and the API answers
404 on the first command	The proxy has no route for the domain — usually a typo in the host rule, or `COMPOSE_FILE` missing from `.env`
An HTML page from the second command	Something else is answering — another app, or another install on this machine
A certificate error	Routing is right; the certificate is missing or still being issued
`Could not resolve host`	DNS

`./up.sh` runs the first check for you every time it starts the stack.

3 Connecting WhatsApp

Attach a WhatsApp Business number. This needs an `https://` address, so finish section 2 first.

Step 1 — collect the values from Meta

At developers.facebook.com → your app → WhatsApp → API Setup:

Value	Where
Phone number ID	API Setup, under the number
WhatsApp Business Account ID	Same screen
App ID	App dashboard header
Access token	The one on API Setup expires in 24 hours — create a System User token in Business Settings for anything permanent

Step 2 — add the number in the app

Settings → WhatsApp → Accounts → Add account. Paste the display number (digits only, no `+` and no spaces), the Phone Number ID, the WABA ID, the App ID and the token. The token is encrypted at rest with AES-256-GCM.

Step 3 — read your webhook verify token

```
cd forge-growth
grep '^META_WEBHOOK_VERIFY_TOKEN=' .env
```

Step 4 — test the handshake before touching Meta

This is the exact request Meta sends when you press Verify and save. Running it yourself turns a slow, silent failure into a one-second answer:

```
cd forge-growth
TOKEN=$(grep '^META_WEBHOOK_VERIFY_TOKEN=' .env | cut -d= -f2-)
curl -s "https://crm.example.com/api/webhook/whatsapp?
hub.mode=subscribe&hub.verify_token=$TOKEN&hub.challenge=42"
42
```

If it prints `42`, Meta will verify successfully. Anything else — fix it here first.

Step 5 — point Meta at it

WhatsApp → Configuration → Webhook → Edit:

Field	Value
Callback URL	`https://crm.example.com/api/webhook/whatsapp`
Verify token	the `META_WEBHOOK_VERIFY_TOKEN` from step 3

Save, then under Webhook fields → Manage, subscribe to **messages**. That single field covers messages, delivery statuses, echoes and template events.

> **The path is /api/webhook/whatsapp**
>
> Not `/api/webhook`. A mismatch gives a silent 404 at verification time and no useful message from Meta.

Step 6 — confirm

- **Settings → WhatsApp → Webhook → Send Test Webhook** → "Incoming text message". Expect an audit row with `processed` and `records_extracted=1`.
- Send a real WhatsApp message to the number. It should appear in Chats.
- Send one from the app to confirm the outbound direction.

> **One callback address per WhatsApp Business Account**
>
> Every phone number under one WABA shares a single callback URL. Pointing it at a new install takes inbound messages away from whatever had it before — instantly and with no warning. If you are moving a live number, plan it as a cutover, not a configuration change.

4 Connecting Instagram

The **Social** tab — Instagram DM automations, AI agents, the inbox and analytics — runs on a Meta app of its own. This also needs an `https://` address.

Step 1 — check the encryption key exists

```
cd forge-growth
grep -c '^SOCIAL_ENCRYPTION_KEY=.' .env
1
```

The installer generates it. If this prints `0` — an install made before 3.1.1 — re-run `./install.sh`, which fills it in and keeps everything else. Without it, the app refuses to save Instagram credentials.

Step 2 — register these addresses in your Meta app

Setting	Value
Instagram OAuth redirect	`https://crm.example.com/api/social/instagram/oauth/callback`
Webhook callback (Instagram)	`https://crm.example.com/api/social/instagram/webhook`
Privacy policy URL	Your own business's privacy policy page

Step 3 — enter the app credentials and connect

Settings → Instagram → Connection → App credentials. Enter the App ID and secret, the Instagram App ID and secret, and a webhook verify token of your choosing — paste the same token into the app's webhook setup. Then connect each Instagram account from the same screen.

Step 4 — turn on syncing

Posts, insights and comments sync in the background, which is off until you switch it on:

```
cd forge-growth
echo 'SOCIAL_JOBS_ENABLED=1' >> .env
./up.sh
```

Optional

- The Social section's AI features use their own OpenRouter key: **Settings → AI models → Instagram AI**.
- People other than admins need the Social page granted: **Settings → Team**.

5 Server, from source

Use this only if you intend to change the code. It builds the images on the machine, which needs about 2 GB of free memory and a few minutes.

```
git clone https://github.com/Forgemind-git/ForgeGrowth-OSS.git
cd ForgeGrowth-OSS
./scripts/install.sh --domain crm.example.com
```

The same flags apply as in section 1. Without a domain:

```
./scripts/install.sh --yes
```

Managing a source install

Command	Does
`./scripts/up.sh`	Start and verify the public URL
`./scripts/up.sh --build`	Rebuild from the checkout, then start
`./scripts/down.sh`	Stop, keeping data
`git pull && ./scripts/install.sh`	Upgrade
`./scripts/uninstall.sh --purge`	Remove, including all data

> **A source install does not upgrade by pulling images**
>
> The images are built locally. Changes reach the running site only when you rebuild. A `git pull` alone moves the files on disk and nothing else, which looks exactly like a feature failing to arrive.

Running the tests

```
cd backend  && REQUIRE_DB=1 npm test && npm run lint
cd frontend && npm run test:unit && npm run lint
```

Without `REQUIRE_DB=1`, the database-backed third of the backend tests quietly skip when no database is reachable.

6 Local install — macOS

Works on both Apple Silicon and Intel. The published images carry `linux/amd64` and `linux/arm64`, so nothing is emulated.

Step 1 — Docker

Install Docker Desktop, OrbStack or Colima, start it, then confirm:

```
docker compose version
```

Step 2 — install

```
cd ~
bash -c "$(curl -fsSL https://raw.githubusercontent.com/Forgemind-git/ForgeGrowth-OSS/
v3.1.1/scripts/install.sh)" -- \
  --version v3.1.1 --yes
```

`--yes` means "localhost, ask me nothing". It creates `~/forge-growth` and serves on port 8080 — or the next free port, which it tells you.

Step 3 — sign in

```
cd ~/forge-growth
grep -E '^(BOOTSTRAP_ADMIN_EMAIL|BOOTSTRAP_ADMIN_PASSWORD)=' .env
```

Open the URL it printed and sign in with those.

Everyday commands

```
cd ~/forge-growth
./up.sh                          # start
./down.sh                        # stop, keeping data
docker compose logs -f backend   # follow the log
open ~/forge-growth              # reveal in Finder (Cmd+Shift+. shows .env)
```

7 Local install — Windows

The installer is a shell script, so it runs inside a Unix shell. WSL2 is the recommended route; Git Bash also works.

Step 1 — install WSL2

In PowerShell as Administrator, then reboot when prompted:

```
wsl --install
```

Open **Ubuntu** from the Start menu and create your Linux user.

Step 2 — Docker Desktop

Install Docker Desktop, then enable Settings → Resources → WSL integration for your distro. Confirm from inside the Ubuntu terminal:

```
docker compose version
```

Step 3 — install, from the Ubuntu terminal

```
cd ~
bash -c "$(curl -fsSL https://raw.githubusercontent.com/Forgemind-git/ForgeGrowth-OSS/
v3.1.1/scripts/install.sh)" -- \
  --version v3.1.1 --yes
```

> **Install into the Linux filesystem, not the Windows drive**
> Use `~/forge-growth`. A location under `/mnt/c/...` is dramatically slower and confuses file-watching. `cd ~` first, every time.

Step 4 — sign in

```
cd ~/forge-growth
grep -E '^(BOOTSTRAP_ADMIN_EMAIL|BOOTSTRAP_ADMIN_PASSWORD)=' .env
```

Open the printed URL in your normal Windows browser — WSL2 forwards `localhost`.

Reaching the files from Windows

Run `explorer.exe .` from the Ubuntu terminal, or type this in Explorer's address bar:

```
\\wsl$\Ubuntu\home\<your-user>\forge-growth
```

8 ngrok — WhatsApp against a laptop

Meta will not accept a `localhost` or plain-HTTP callback, so a laptop needs a public HTTPS address borrowed from a tunnel.

> **Use a Meta test number, never a live one**
> Every Meta app includes a free test number on its own WhatsApp Business Account, so its callback is independent of any production number. A laptop also sleeps, and inbound messages arriving then are lost rather than queued.

Step 1 — install and authenticate

```
brew install ngrok                      # macOS; on Windows install it inside WSL2
ngrok config add-authtoken <your-token>  # dashboard.ngrok.com → Your Authtoken
```

Step 2 — claim a static domain

dashboard.ngrok.com → Domains → New Domain. The free tier includes one. Without it the URL changes on every restart and the Meta webhook has to be re-pointed each time.

Step 3 — start the tunnel

Use the port your install actually reported, and leave that terminal running:

```
ngrok http --url=your-name.ngrok-free.app 8080
```

Step 4 — confirm the tunnel reaches the app

```
curl -s -o /dev/null -w '%{http_code}\n' https://your-name.ngrok-free.app/
200
```

Step 5 — run Meta's handshake yourself

```
cd ~/forge-growth
TOKEN=$(grep '^META_WEBHOOK_VERIFY_TOKEN=' .env | cut -d= -f2-)
curl -s "https://your-name.ngrok-free.app/api/webhook/whatsapp?
hub.mode=subscribe&hub.verify_token=$TOKEN&hub.challenge=42"
42
```

This works before any WhatsApp account is configured — the check falls back to the token in `.env` — which separates "my tunnel is wrong" from "my credentials are wrong".

Step 6 — keep using localhost for the interface

The tunnel is for Meta. Keep opening the app at the `localhost` address the installer printed.

Step 7 — configure Meta and test

Field	Value
Callback URL	`https://your-name.ngrok-free.app/api/webhook/whatsapp`
Verify token	the value from step 5
Subscribe to	`messages`

Add your own phone under API Setup → To → Manage phone number list; a test number only messages people on that list. Then watch every delivery arrive in ngrok's inspector at `http://127.0.0.1:4040` — it shows each request Meta makes with the full body, and lets you replay them.

> **Things that will bite**
> Closing the ngrok terminal kills the tunnel silently. The token on Meta's API Setup screen expires after 24 hours — when inbound stops for no apparent reason, this is usually why.

9 Upgrading, backup, removal

Upgrading

Re-running the installer is the upgrade. From the install directory:

```
cd forge-growth
./install.sh                    # stays on the version it is pinned to
./install.sh --version v3.1.1   # move deliberately, and stay there
```

It re-downloads the compose file and scripts, pulls the images, applies any new database migrations at start-up, and restarts. `.env` is never overwritten, and secrets are kept. For a source install: `git pull && ./scripts/install.sh`.

> **Downgrading is not supported**
> Migrations run forward at start-up, so pinning back to an older release runs old code against a newer schema. Take a dump first.

Backup — the things you cannot recreate

```
cd forge-growth
grep -E '^(FORGECRM_ENCRYPTION_KEY|SOCIAL_ENCRYPTION_KEY|POSTGRES_PASSWORD)=' .env
```

> **Store these off the server, with the database**
> The two encryption keys are the only things that decrypt stored Meta, Google and Instagram credentials. `POSTGRES_PASSWORD` is applied only when the database is first created, so a regenerated `.env` cannot open an existing database. Lose any of them and the failure names the wrong thing — an authentication error that reads like a network problem.

Database dump

```
cd forge-growth
docker compose exec -T postgres pg_dump -U forgegrowth forgegrowth \
  | gzip > backup-$(date +%F).sql.gz
```

Removal

```
cd forge-growth
docker compose down -v        # stops and DELETES all data
cd .. && rm -rf forge-growth
```

10 When it goes wrong

"pull access denied for minio/minio"

An install from before 3.1.0. The official MinIO image has been removed from Docker Hub; 3.1.0 and later use a maintained build of the same server. Move to a current release — the Media Library is kept:

```
./install.sh --version v3.1.1
```

"no matching manifest for linux/arm64/v8"

An image with no build for your processor. Compose pulls concurrently and reports one failure against every service, so the message names `postgres`, `redis`, `minio` and `caddy` — all innocent. Check the two images this project publishes:

```
docker manifest inspect ghcr.io/forgemind-git/forgegrowth-web:latest | grep architecture
```

To carry on regardless, run everything under emulation: `export DOCKER_DEFAULT_PLATFORM=linux/amd64`.

"Secrets cannot be saved yet: the server has no SOCIAL_ENCRYPTION_KEY"

Instagram on an install made before 3.1.1. Re-run `./install.sh`; it generates the key and changes nothing else.

It logs me out on every refresh, or says Unauthorized when I change tab

The login cookie is marked `Secure` and the site is being served over `http://`, so the browser silently discards it. Reach the site over `https://`. If HTTPS is terminated by a proxy this stack cannot see, set the flag explicitly:

```
cd forge-growth
echo 'COOKIE_SECURE=true' >> .env
./up.sh
```

Every container is healthy but the site returns 404

Nothing is routing the domain to the install. From inside the stack nothing is wrong, which is why container health cannot detect it. Check that `COMPOSE_FILE` is in `.env`, then run the checks in section 2c.

"port 8080 is in use"

Not an error — the installer moved to the next free port and told you which. Find what holds a port:

```
lsof -nP -iTCP:8080 -sTCP:LISTEN     # macOS
ss -lntp | grep 8080                 # Linux
```

A container keeps restarting

Usually a memory ceiling set too low — a container that reaches its limit is killed and restarted, which looks like a container that never finishes starting.

```
cd forge-growth
docker compose ps
docker compose logs --tail 50 backend
```

Raise the ceiling in `.env` — `BACKEND_MEM_LIMIT` first; a busy install wants `1g`.

A second install adopted the first one's database

Installs on one machine are separated by their project name and nothing else. Check which directory owns which:

```
docker compose ls
grep '^COMPOSE_PROJECT_NAME=' forge-growth/.env
```

`up.sh` refuses to start when another directory already owns this install's name — the reason to use it rather than raw Compose.

Useful one-liners

```
cd forge-growth
grep '^BOOTSTRAP_ADMIN_PASSWORD=' .env    # admin password
grep '^META_WEBHOOK_VERIFY_TOKEN=' .env   # WhatsApp webhook verify token
grep '^TLS_MODE=' .env                    # caddy = owns 80/443, proxy = does not
grep '^WEB_PORT=' .env                    # which port the UI is on
docker compose ps                         # container states
docker compose logs -f backend            # follow the backend
docker compose ls                         # every install on this machine
docker volume ls | grep forgegrowth       # where the data lives
```